

LEXINGTON, NE, USA

WMO: 725624

Lat: 40.789N

Lon: 99.771W

Elev: 733

StdP: 92.82

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 14994

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.2	-15.8	-22.3	0.6	-17.5	-19.6	0.7	-14.6	14.6	2.8	12.8	1.6	3.2	300	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	35.1	22.1	33.0	21.9	31.8	21.8	24.7	30.8	23.8	30.0	22.9	29.3	5.9	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.8	19.2	28.4	22.1	18.4	27.8	21.1	17.2	26.8	79.1	31.1	75.5	29.7	71.9	29.5	28.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.1	11.5	10.0	-24.0	38.1	2.9	2.2	-26.2	39.6	-27.9	40.9	-29.5	42.1	-31.7	43.7		
(5)			-24.4	26.5	2.9	1.1	-26.5	27.2	-28.1	27.9	-29.7	28.5	-31.8	29.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.5	-2.6	-1.2	4.7	9.6	15.4	21.3	24.2	22.6	18.4	10.6	4.1	-2.0		
	DBStd	10.72	6.09	6.37	6.07	5.19	4.70	3.82	3.06	3.14	4.63	5.13	5.19	5.67		
	HDD10.0	1574	389	314	182	68	8	0	0	0	2	53	184	373		
	HDD18.3	3378	648	547	423	264	117	15	2	4	57	243	426	631		
	CDD10.0	1752	0	1	17	56	174	339	441	390	253	72	8	0		
	CDD18.3	514	0	0	0	2	24	104	184	137	58	4	0	0		
	CDH23.3	5643	0	1	21	74	351	1161	1972	1298	672	88	5	0		
	CDH26.7	2327	0	0	3	16	119	472	901	532	263	21	0	0		
(14) Wind	WSAvg	4.3	4.1	4.5	5.0	5.6	5.0	4.5	3.7	3.4	4.0	4.2	4.2	4.1		
(15) Precipitation	PrecAvg	590	11	15	29	66	91	97	80	75	46	45	21	13		
	PrecMax	868	36	37	72	179	257	207	238	142	109	149	85	78		
	PrecMin	271	0	1	0	16	21	18	3	21	7	0	0	0		
	PrecStd	120	8	9	21	37	54	48	54	33	26	36	19	15		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	20.2	26.0	28.7	32.8	36.2	37.5	37.0	35.0	29.1	23.7	16.2		
		MCWB	7.4	9.7	12.9	14.7	17.8	21.0	22.0	21.9	20.0	16.2	11.9	7.3		
	2%	DB	12.4	15.2	22.3	25.5	29.9	33.7	35.2	33.7	32.3	26.0	19.9	12.5		
		MCWB	5.5	6.9	11.5	14.2	18.2	21.4	23.0	22.5	20.1	15.2	10.6	6.1		
	5%	DB	8.9	11.8	18.6	22.5	27.4	31.8	33.0	31.8	30.0	22.8	16.4	9.0		
		MCWB	3.9	5.7	10.1	13.2	17.1	21.4	22.9	22.6	19.5	13.9	8.8	4.1		
	10%	DB	6.0	8.0	15.2	19.6	24.8	29.3	31.8	29.8	27.4	20.1	13.0	6.2		
		MCWB	2.2	3.8	8.5	11.9	16.5	20.4	22.8	22.1	18.2	12.8	7.5	2.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	10.2	15.5	17.1	21.5	24.8	26.3	25.5	23.0	18.8	13.7	9.4		
		MCDB	15.7	18.0	20.2	24.3	28.0	30.6	32.3	31.4	30.1	25.2	19.2	15.1		
	2%	WB	5.6	7.8	13.0	15.3	19.9	23.2	25.0	24.4	21.7	16.8	11.4	6.1		
		MCDB	11.7	14.2	19.4	22.7	26.1	30.2	31.3	30.3	28.9	22.2	17.6	12.1		
	5%	WB	4.0	5.8	10.8	13.8	18.5	22.3	24.1	23.5	20.6	15.3	9.7	4.2		
		MCDB	8.2	11.2	17.7	20.8	25.0	29.3	30.4	29.3	27.1	20.7	16.0	8.6		
	10%	WB	2.4	4.0	8.9	12.5	17.3	21.2	23.3	22.6	19.6	13.8	7.7	2.5		
		MCDB	5.7	8.2	15.4	18.7	23.1	28.0	29.8	28.2	25.5	19.4	13.5	6.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	12.5	12.5	14.1	13.7	13.2	12.9	12.7	12.5	14.5	14.4	14.3	12.3		
		MCDBR	17.5	18.5	21.0	19.4	17.1	16.0	15.4	15.4	17.2	19.3	19.9	17.3		
	5% WB	MCWBR	11.2	11.3	11.4	10.0	7.8	6.7	6.6	6.5	7.2	9.7	11.4	11.3		
		MCWBR	16.4	17.3	18.5	16.8	14.2	13.8	13.1	13.2	14.4	15.1	18.7	16.2		
(40) Clear-Sky Solar Irradiance	taub		0.269	0.273	0.298	0.328	0.358	0.375	0.391	0.384	0.352	0.311	0.274	0.264		
		taud	2.507	2.508	2.466	2.402	2.395	2.382	2.377	2.384	2.427	2.519	2.540	2.514		
	Ebn at Noon		907	954	959	944	917	898	880	878	886	895	894	884		
		Edh at Noon	73	84	99	114	117	119	119	115	103	83	70	67		
(44) All-Sky Solar Radiation	RadAvg		2.27	3.12	4.25	5.30	6.13	7.04	6.94	6.02	4.88	3.49	2.50	1.96		
	RadStd		0.20	0.22	0.35	0.29	0.41	0.42	0.27	0.30	0.28	0.35	0.17	0.16		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (18 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page